

Hao (August) Xu

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Education

Doctor of Philosophy

Aug 2024 - Now

- **Deakin University Joint Program with Paralympics Australia**
- **Supervisor: Associate Prof. Sunil Aryal**
- Research Area: Computer vision and machine learning for video understanding, including temporal modelling, event detection, object tracking, self/semi-supervised learning, and multimodal representation learning.

Master of IT

Graduate 2024

- **The University of Melbourne** (Major in Distributed Computing)
- Graduated With Distinction

Bachelor of Science

Graduate 2022

- **The University of Melbourne** (Major in Computing and Software Systems)

Experience

PhD Researcher Intern

Dolby Laboratories – Sydney

May 2026 – Sep 2026

- Developed multimodal video representation learning methods for long-form temporal video understanding.
- Conducted end-to-end research from problem formulation and model development to large-scale experimentation and evaluation.
- Research led to an **ACM Multimedia workshop** publication and an **internal IP** submission, with work submitted to **ICLR**.

PhD Candidate

Paralympics Australia – Table Tennis Australia

Aug 2024 - Now

- Led applied computer vision research on temporal video understanding, event detection, and object tracking in real-world video.
- Designed and trained deep learning models for multimodal event detection and occlusion-aware tracking.
- Developed technical architecture and standardized data labeling protocols for large-scale video annotation.
- Coordinated between academia and industry to align project goals and stakeholder expectations.

Research and Development Engineer (Intern)

July 2023 – Oct 2023

Agriculture Victoria Research, Melbourne

- Developed a full-stack web application using JavaScript and Python Flask, improving interface usability and responsiveness.
- Deployed the application on Azure and integrated cloud APIs to automate data mapping and streamline workflows.

Software Engineer (Intern)

Dec 2022 - Feb 2023

NIO, Shanghai

- Developed automated testing tools using Pytest, improving testing efficiency by 10%.
- Collaborated across engineering and technical support teams to identify and resolve software issues.

Selected Publications

Temporal Feature Distillation for Label-Efficient Precise Event Spotting

ACM Multimedia (ACM MM) · 2026

- Proposed Temporal Feature Distillation, a semi-supervised framework for learning temporally consistent video representations under strong/weak augmentations.
- Demonstrated significant gains in low-label regimes by combining TFD with motion-aware augmentations, improving temporal localization accuracy and outperforming prior SSL approaches that fail to retain fine-grained motion cues.

Multi-Focus Temporal Shifting for Precise Event Spotting

IEEE International Conference on Multimedia and Expo (ICME) · 2026

- Proposed a Multi-Focus Temporal Shifting module that enhances lightweight PES models with multi-scale temporal shifts and grouped spatial attention, enabling better modelling of long-range dependencies and salient regions with minimal overhead (+1.3% params, efficient GFLOPs) .
- Achieved state-of-the-art performance among lightweight temporal models across five video benchmarks, improving mAP by up to 4.09 points while adding only 1.3% parameters.

TOTNet: Occlusion-Aware Temporal Tracking for Robust Ball Detection in Sports Videos

Computer Vision and Image Understanding (CVIU) · 2026

- Developed an occlusion-aware temporal tracking framework for robust small-object tracking under severe visibility degradation.
- Combined 3D/2D CNNs with temporal modelling to recover ball trajectories through full occlusions. Introduced an occlusion-focused evaluation protocol and augmentation strategy that improved robustness and generalization.

Audio for Sports Highlight Detection: A Comparative Empirical Study

ACM Multimedia Workshop on Multimedia Content Analysis in Sports (MMSports) · 2026

- Conducted a systematic empirical study of audio for long-form sports highlight detection, demonstrating that a lightweight audio-only temporal model can outperform several existing audio-visual approaches.
- Showed that temporal modelling and complementary audio-visual cues are critical for highlight detection; audio-visual fusion achieved **46.92 mAP**, while source-separated analysis revealed that commentary/vocal and background event sounds provide complementary information.

Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges

IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI) · Under Review

- Proposed a new taxonomy summarizing methods and datasets for action/event spotting.
- Outlined challenges and future directions in the field.

Technical Skills

- **AI/ML & Data Science:** Python, PyTorch, TensorFlow, SQL, data preprocessing & model optimization, OpenCV
- **Software Development:** Java, C++, C, JavaScript, ReactJS, Spring Boot, JUnit, Pytest
- **Cloud & Tools:** Azure, AWS, GCP, MongoDB, MySQL, PostgreSQL, Linux/Windows

Certifications

- Microsoft Azure Fundamentals ([Link](#))